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ORIGINAL RESEARCH ARTICLE

Effect of Double-Wire Cold Feeding on Wire Arc Additive Manufacturing of SS309L Layer Geometry

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Wire arc additive manufacturing (WAAM) offers significant advantages for the fabrication of stainless-steel components; yet, the optimization of process parameters remains critical for achieving desirable geometric and microstructural properties. This study investigates the effects of travel speed, welding current, and wire feed ratio on the layer geometry and microstructure of SS309L stainless steel produced via robotic double-wire MIG welding. Using a Taguchi L9 orthogonal array, the main and interaction effects of these parameters were systematically evaluated. Analysis of variance revealed that wire feed ratio is the most influential factor, contributing (43.6%) to wall height variation, followed by travel speed (22.3%) and welding current (15.9%). A linear regression model was developed to predict wall height, achieving strong agreement with experimental results, with a maximum wall height of 30.79 mm and a deviation below 2.5%. Microstructural analysis showed that the incorporation of cold wire feeding promotes the formation of vertically aligned dendritic structures and a refined granular matrix, resulting in dense, defect-free deposits. In contrast, specimens produced without cold wire feeding exhibited coarser dendritic features and less uniform morphology. These improvements in microstructure are directly associated with enhanced surface quality and mechanical integrity of the additively manufactured walls. The results provide valuable guidance for tailoring WAAM process parameters to achieve precise geometries and consistent material properties in stainless-steel components, thereby informing continuous optimization efforts.

Keywords double-wire MIG welding, layer-by-layer material deposition, wire arc additive manufacturing (WAAM)

1. Introduction

Additive manufacturing (AM) has rapidly evolved into a leading method for fabricating components with mechanical properties that rival or surpass those produced through traditional techniques such as casting, forging, or machining (Ref 1). As industries push for faster production cycles, AM is shifting from a prototyping tool to a robust solution for full-scale manufacturing. Metal AM technologies are typically categorized by their energy source: electron beam (Ref 1-3), laser beam (Ref 4-6), and electric wire arc systems (Ref 7-15). According to Santos et al., electron and laser-based methods offer excellent surface quality due to their concentrated energy input; however, they suffer from slower build rates, higher operational costs, and extended processing durations (Ref 16). In contrast, wire arc additive manufacturing (WAAM), particularly through direct metal deposition (DMD), provides faster deposition and more economical equipment, though it struggles with precision and geometric consistency. DMD has been successfully applied to materials such as titanium alloys (Ref 17-19), stainless steels (Ref 20-24), and Inconel superalloys (Ref 25-27).

WAAM often utilizes metal inert gas (MIG) welding to emulate DMD, layering molten wire onto a substrate to build structures incrementally. Stainless-steel SS309L is especially effective for vertical deposition tasks and multi-positional builds, provided the operating temperature remains below approximately 399 °C (Ref 28). Achieving consistent surface finish and dimensional accuracy requires meticulous control of process variables (Ref 29). Depending on the application, wall thickness may vary; thicker walls improve deposition efficiency (Ref 30), while thinner walls are preferred in weight-sensitive sectors like aerospace and automotive (Ref 31).

Recent research has emphasized improving weld bead characteristics and enhancing process reliability. Xiong et al. demonstrated that dynamically adjusting wire feed speed can significantly improve bead uniformity and dimensional control (Ref 32). Xiang et al. explored how the torch-to-substrate standoff distance affects process stability (Ref 33), while Yang et al. developed predictive models for bead geometry in robotic GMAW-based rapid manufacturing (Ref 34). To achieve optimal weld profiles and reduce defects such as porosity and cracking, comprehensive parameter optimization is essential (Ref 35, 36).

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One of the key challenges limiting broader adoption of metal AM is the incomplete understanding of fatigue performance. Kumar et al. (Ref 37) investigated how factors like material hardness, tensile strength, and defect size influence fatigue life, proposing simplified methods to estimate S–N curves using readily measurable properties. Their approach offers a practical framework for evaluating fatigue behavior in AM components. Gas metal arc welding (GMAW)-based wire arc additive manufacturing of stainless steel requires precise control of bead geometry to ensure dimensional accuracy and mechanical integrity. Chaudhari et al. conducted a parametric study analyzing factors such as current, voltage, and travel speed, and their influence on bead shape and quality. The results highlight optimal process parameters that minimize defects and improve surface finish, which are critical for reliable additive manufacturing applications (Ref 38).

Although E309LTI-1/T1-4 shows promise for industrial applications, its performance in additive manufacturing remains underexplored, particularly regarding optimal welding conditions.

This study introduces a novel process variable (wire feed ratio) defined as the ratio between the primary MIG wire feed speed and that of an auxiliary feeder. The research aims to optimize three critical parameters in the double-wire MIG surfacing-metal deposition (DW-MIG-SMD) process: travel speed (TS), welding current (A), and wire feed ratio (WR). The primary objective is to enhance wall height formation while minimizing bead discontinuities, ultimately improving the structural integrity and dimensional precision of the deposited walls.

2. Experimental Setup

2.1 Experimental Plan and Material Selection

Figure 1 illustrates the WAAM setup, incorporating a robotic MIG torch and a computer-controlled double-wire deposition machine (CADWDM). Depositions were performed on St 3 low-carbon steel plates (150 × 100 × 8 mm) using a 1.2 mm diameter SHIELD-BRIGHT 309L stainless-steel wire. Table 1 summarizes the chemical composition of ASTM standards (ASTM A240/AWS A5.9 ER309L for SS309L filler wire, ASTM A283 Grade C for substrate steel).

2.2 Parameter Selection

To define suitable ranges for Taguchi's orthogonal array, process SS309L welding parameters were selected based on prior studies. Preliminary trials helped establish a practical operating window for double-wire MIG deposition. The setup included a welding current of 160 A, main wire feed speed of 3 m/min, and synchronized cold wire feed. Travel speed was set at 2 mm/s, with the torch positioned vertically (90°), cold wire angled between 30° and 60°, a stick-out of 15 mm, and arc length maintained between 1 and 2 mm for stable deposition.

The welding process initially exhibited several defects, particularly within the first 50 mm of deposition, characterized by fluctuating, droplet formation, and swirling bead patterns. Beyond this distance, the cold wire feed (CWF) deviated from the intended path. This deviation was attributed to the angle and proximity of the cold wire feed relative to the main welding wire. When the cold feed wire made contact with the main wire outside the welding gun, welding defects occurred, resulting in the CWF inclining away from the workpiece during deposition, as illustrated in Fig. 2.

Initial deposition was stable, but instability appeared after 50 mm of build height, characterized by droplet ejection and swirling bead patterns. This was caused by the cold wire deviating due to excessive inclination and close proximity to the main wire, leading to electrode contact and bead defects. The issue was resolved by adjusting the cold wire angle to 25°-55° and maintaining a 2-3-mm gap between wire tips during arc formation to prevent premature melting. Subsequent trials showed that high travel speeds combined with low current resulted in insufficient melting (Fig. 3a), while high current with low speed caused overheating (Fig. 3b). The most consistent and smooth bead profiles were achieved by increasing the wire feed ratio (WR), as shown in Fig. 3(c). Optimal deposition occurred at 190-210 A welding current, 3-4 mm/s travel speed, and WR between 1.5 and 2.5. These results established the effective parameter range, detailed in Table 2, for experimental planning.

3. Results and Discussions

3.1 Taguchi Methods

Minitab 18 was used to analyze experimental data and define suitable welding parameter ranges for orthogonal array design. The aim of using the Taguchi method is to identify the dominant main effects. This software supports design of experiments tools like regression, ANOVA, and main-effect plots, enabling precise evaluation of how travel speed (TS), welding current, and wire ratio (WR) affect bead formation. These variables were varied systematically using an L9 orthogonal array to study their individual and combined effects. The main response measured was bead width, reflecting deposition quality and consistency. Minitab's statistical tools helped identify the optimal parameter window that balances geometry, efficiency, and process stability. The Taguchi method minimized the number of trials while maximizing insight, allowing thorough analysis with just nine experiments. All calculations, including response tables and variance analysis, were performed using Minitab to ensure accuracy. The resulting bead-width data for each condition are presented in Table 3, forming the basis for selecting the most effective welding parameters for the double-wire MIG deposition process.

3.1.1 Height Wall Construction. The ANOVA test is conducted at a 95% confidence level to find the variable that shows the highest. At a 95% confidence level, an ANOVA test was performed to identify the most statistically significant input parameter affecting weld bead height. The results, summarized in Table 4, indicate that wire feed ratio (WR) is the most dominant factor, followed by travel speed (TS), with welding current (I) showing the least influence. The model achieved an R-squared value of 81.73% and an adjusted R-squared of 70.84%, confirming a strong fit between the input variables and the response.

Significance was determined using a p value threshold of $\leq$ 0.05. WR showed the highest significance, as illustrated in Fig. 4, where increased cold wire feed directly correlated with greater wall height due to enhanced deposition. WR contributed

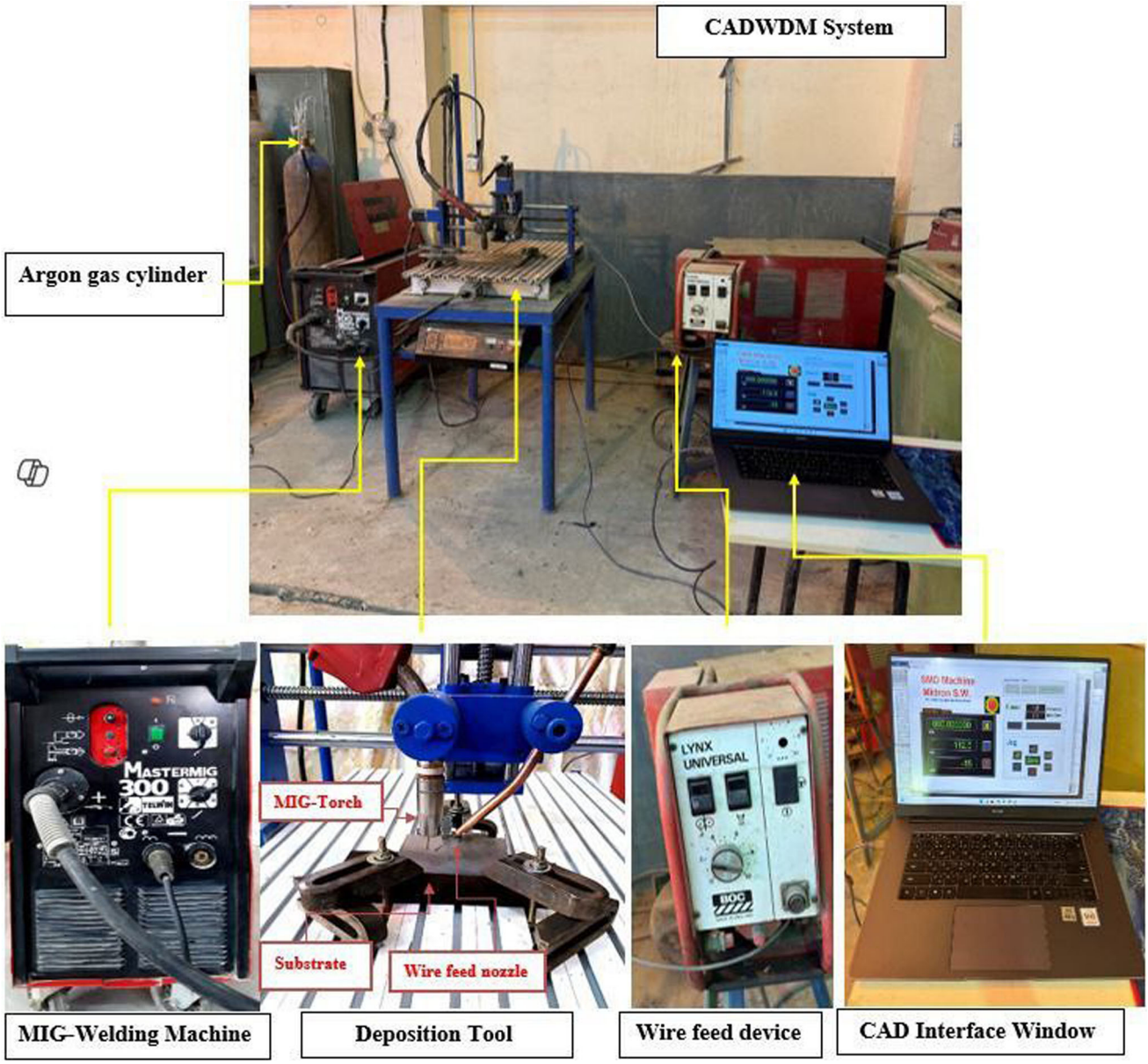


Fig. 1 The CADWDM system primary units

Table 1 Chemical composition of wire and substrate compared with ASTM standards in wt.%

Element	Wire	ASTM/AWS ER309L range	Substrate	ASTM A283 grade C range
C	0.03%	≤ 0.03%	0.1%	≤ 0.1%
Si	0.9%	≤ 0.75%	0.03-0.15%	0.15-0.40%
Mn	1.3%	1.0-2.5%	0.2-0.45%	0.2-1.3%
P	…	…	0.04%	≤ 0.035%
S	…	…	0.04%	≤ 0.04%
Mo	0.1%	≤ 0.75%	…	…
Ni	12.5%	12.0-14.0%	…	…
Cr	24%	23.0-25.0%	…	…
Cu	0.1%	≤ 0.75%	…	…
Fe	REM.	Balance	REM.	Balance



Fig. 2 Deviation of the cold wire feed during the welding process

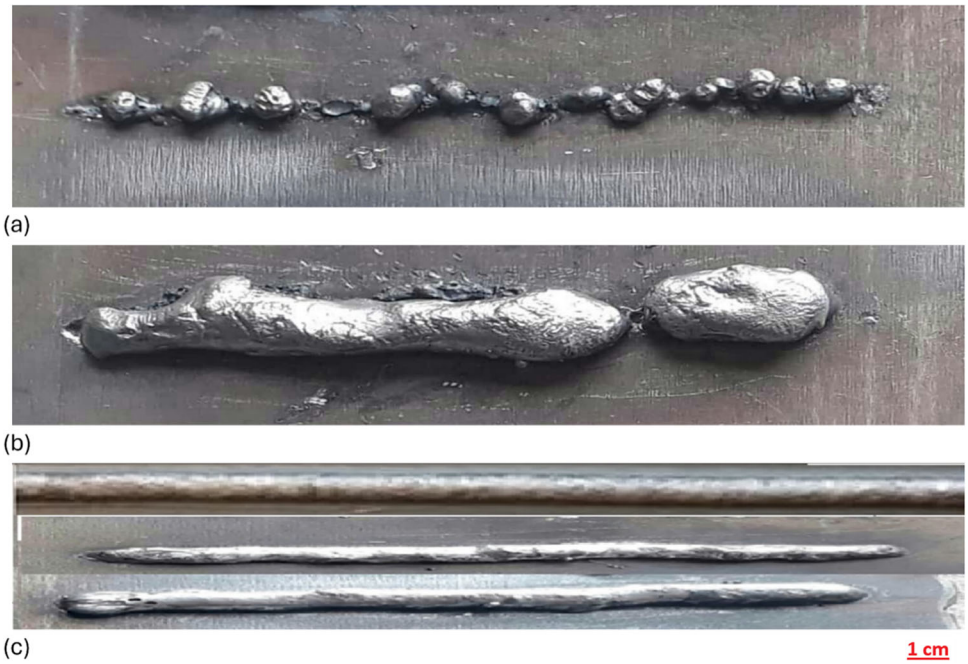


Fig. 3 Drawbacks of melting deposits: (a) insufficient melting deposits, (b) excessive melting deposits, and (c) stable and smooth bead deposits

Table 2 Process parameters and levels

Input parameter (symbol)	Units	Levels		
		1	2	3
Welding current (I)	(A)	160	190	210
Travel speed (TS)	(mm/sec)	2	3	4
WIRE Ratio (WR) ($W.F_1^{*}$ /$W.\ F_2^{**}$)	(m/min)	1.5	2	2.5

* Wire feed comes from the welding machine.
** Cold wire is supplied through an external nozzle at a constant feed rate of 2 m/min.

Table 3 Experimental conditions for the L9 orthogonal array design

Exp. No	I, A	TS, mm/s	WR, m/min	Hav, mm	S/N Ratio of Hav
1	160	2	1.5	27.00	28.6
2	160	3	2.0	28.00	28.9
3	160	4	2.5	29.50	29.3
4	190	2	2.0	26.90	28.5
5	190	3	2.5	29.00	29.2
6	190	4	1.5	28.00	28.9
7	210	2	2.5	27.50	28.7
8	210	3	1.5	25.03	27.9
9	210	4	2.0	28.17	28.9

Table 4 ANOVA for means of total height

Source	DOF	Seq. SS	Adj. MS	*F* value	*P* value	Percent contribution
I	1	2.163	2.163	4.36	0.091	15.88%
TS	1	3.039	3.0388	6.12	0.056	22.30%
WR	1	5.940	5.9401	11.96	0.018	43.60%
Error	3	2.483	0.4966			18.22%
Total	6	13.625				100.00%
R-Sq. = 81.78%, *R*-Sq (adj) = 70.84%						

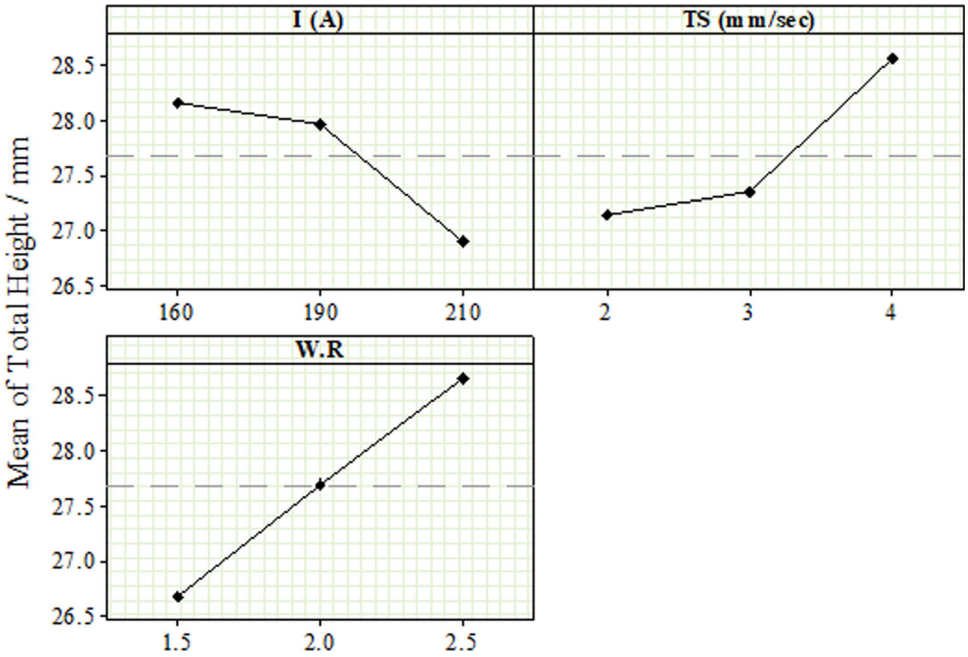


Fig. 4 Effects of main process parameters on the total wall height

approximately 43.60% to the variation in height, TS contributed 22.30%, and welding current had minimal effect.

A regression model was developed using data from Table 4 and the Taguchi L9 orthogonal array to quantify the relationship between bead height and input parameters. Trends across the parameter sweeps revealed that increasing welding current from 160 to 210 A reduced bead height from 28.2 to 26.8 mm, likely due to excessive heat flattening the weld pool. Conversely, raising travel speed from 2 to 4 mm/s increased height from 26.7 to 28.3 mm by limiting arc energy per unit length. Similarly, increasing WR from 1.5 to 2.5 mm/s raised height from 27.0 to 28.4 mm by supplying more filler material.

The choice of linearity was based on two considerations: (i) the narrow parameter window investigated (160-210 A, 2-4 mm/s, WR 1.5-2.5), within which bead height trends were monotonic and approximately linear; and (ii) the need for a simple, interpretable model linking deposition geometry to controllable inputs. Nonlinear fits (quadratic and polynomial)

were evaluated but did not significantly improve R^2 beyond 83%, indicating limited curvature in the studied range. Physically, higher current spreads the molten pool, reducing height, while increased travel speed and wire ratio promote buildup, both showing near-linear responses in this window. We acknowledge that outside this range, nonlinearities and interactions may become significant, and future work will employ response surface methodology to capture such effects.

These findings confirm that bead geometry is governed by a balance between arc energy and deposition rate. WR has the strongest positive effect, followed by TS, while excessive current reduces wall height. Optimizing these parameters is essential for achieving consistent, high-quality weld profiles in additive manufacturing.

3.1.2 Signal-to-Noise (S/N) Ratio (Larger is Better). The impact of process parameters on the height of additively deposited weld beads is summarized in Table 5, while Fig. 5 illustrates the main effects plots for both mean values and signal-to-noise (S/N) ratios under the "larger is better" criterion. These visualizations help identify the optimal settings for maximizing bead height. The highest wall formation was achieved at a welding current of 160 A, a travel speed of 4 mm/s, and a wire feed ratio (WR) of 2.5.

The S/N ratio plot reveals that robustness decreases as current increases from 160 A to 210 A, while it improves consistently with higher travel speeds (2-4 mm/s) and wire feed ratios (1.5-2.5). Since higher S/N ratios indicate more stable performance, the most consistent bead height results from low current and high travel speed and WR. The steep slope of the WR curve suggests that it has the strongest positive influence on process stability, followed by travel speed, while excessive current negatively affects robustness.

These findings recommend using 160 A, 4 mm/s, and WR of 2.5 to ensure optimal deposition consistency. Further statistical validation through ANOVA or confirmation trials is advised to verify the influence of each factor.

A regression model was developed using the design of experiments methodology to relate welding current, travel speed, and wire feed ratio to bead height. This equation enables prediction of weld bead height for any parameter combination within the tested range, offering a reliable tool for optimizing wall formation in additive manufacturing.

$$H_t = 26.02 - 0.0239I + 0.712\text{TS} + 1.990\text{WR}$$

3.2 Contour Plots for Height

Wall height in additive deposition is strongly influenced by wire feed ratio, which depends on the cold wire feed rate. Increasing the cold wire feed enhances wall height, while higher heat input—typically from lower travel speeds—also contributes to taller weld beads. Figure 6 presents contour plots that visually demonstrate how input parameters affect wall height and help define suitable ranges for industrial applications.

The TS–WR plot shows wall heights below 26 mm at low travel speeds (2 mm/s) and wire ratios (1.5), rising to over 29 mm at TS $\approx$ 4 mm/s and WR $\approx$ 2.5. In the I–WR plot, maximum heights above 29 mm occur only at low currents (160-170 A) and high wire ratios, while currents above 200 A restrict height to under 27 mm. The TS–I plot confirms that high travel speeds improve wall height only when current remains low.

Combining insights from all three plots, the optimal parameter window for maximum wall height is identified: travel speed near 4 mm/s, wire ratio around 2.5, and welding current between 160 and 170 A. These findings highlight that while travel speed and wire feed enhance deposition, excessive current suppresses bead growth. This guidance supports targeted confirmation trials and efficient process optimization.

Table 5 Response for signal-to-noise ratios of total height

Level	I, A	TS, mm/sec	WR, m/min
1	28.99	28.67	28.51
2	28.93	28.72	28.82
3	28.58	29.11	29.14
Delta	0.41	0.44	0.63
Rank	3	2	1

3.3 Confirmation Tests

The maximum weld bead height for wall construction was obtained using a welding current of 160 A, a travel speed of 4 mm/sec, and a wire feed rate of 2.5. This experiment, included in Table 3 L9 design, was repeated to confirm its effectiveness. The resulting bead height was 30.79 mm, closely matching the predicted value of 30 mm. The error percentage was 2.5%, indicating strong consistency between experimental results and predicted outcomes.

3.4 Micro Structural Analysis

The microstructural characterization of the deposited specimens revealed distinct grain growth within the fabricated walls, accompanied by an overall refinement in surface morphology. As illustrated in Fig. 7, the walls produced without cold feeding exhibited a predominantly fine granular matrix interspersed with coarser dendritic formations (Fig. 7a). Conversely, the incorporation of cold feeding promoted the development of vertically aligned dendritic colonies in conjunction with a uniformly smooth granular texture (Fig. 7b–d). Comparable microstructural features have been documented in prior investigations by Chaudhari, Rakesh, et al. (Ref 39).

3.5 Additively Deposited Layer Height

The concept of additively deposited layer height is central to understanding the geometric accuracy and mechanical reliability of WAAM-fabricated stainless-steel components. In the present study, layer height was treated as the primary response variable, since it directly reflects the efficiency of material buildup, the stability of bead formation, and the dimensional precision of the constructed walls. Chaudhari, Rakesh, et al. (Ref 39) observed that bead height scales positively with both wire feed and travel speed, underscoring the critical role of deposition rate and travel dynamics in controlling weld geometry. As shown in the Taguchi L9 orthogonal array experiments, the average wall height varied between 25.03 and 29.50 mm depending on the combination of welding current, travel speed, and wire feed ratio (Table 3). Subsequent ANOVA analysis confirmed that wire feed ratio (WR) was the most influential factor, contributing 43.6% to height variation,

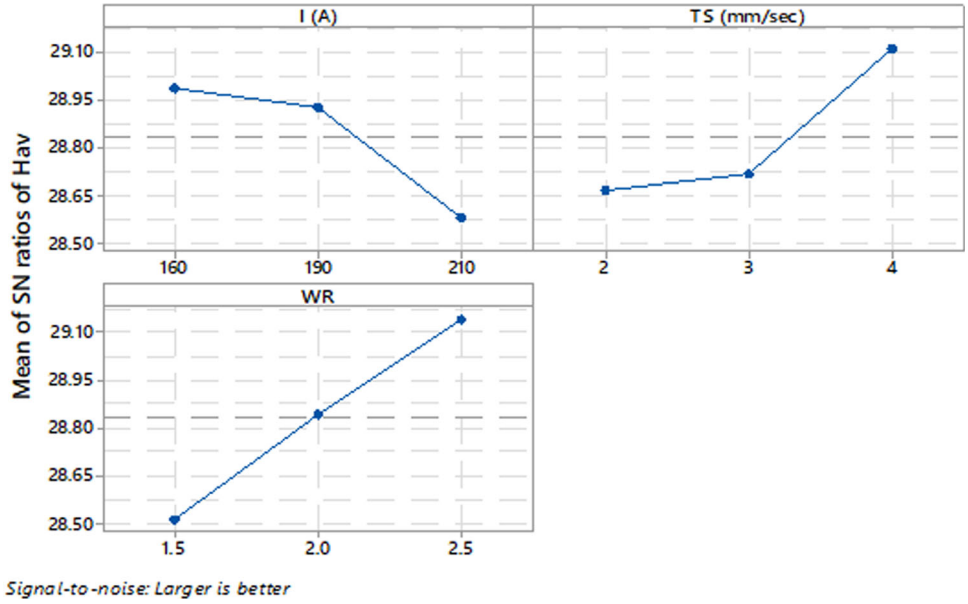


Fig. 5 Main effects plot for SN ratios

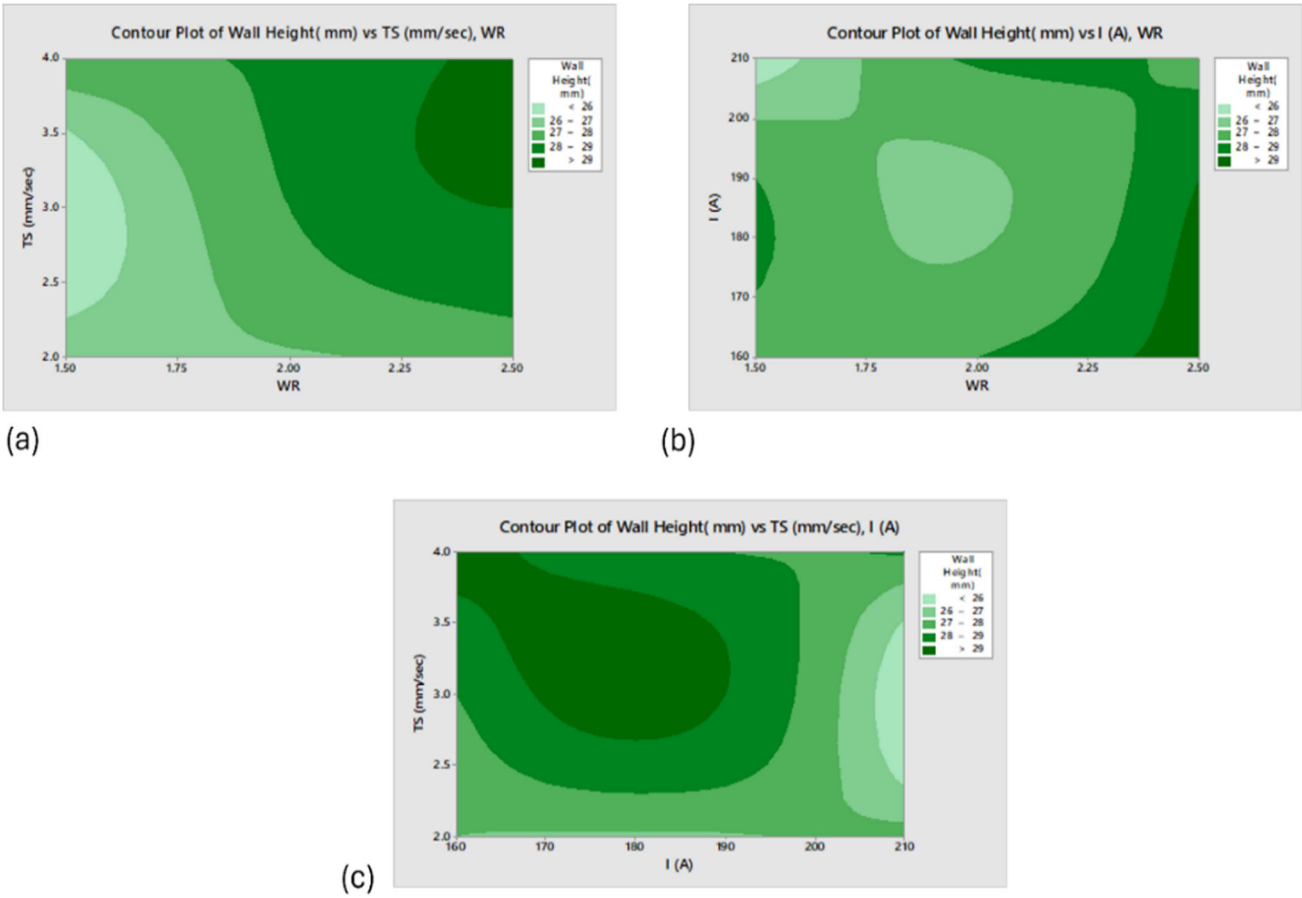


Fig. 6 Contour plot for with respect to width Vs (a) TS (mm/sec) and WR, (b) welding current (A) and WR, and (c) Ts (m/ min) and welding current (A)

followed by travel speed (22.3%) and welding current (15.9%) (Table 4). This dominance of WR highlights the importance of filler supply in governing vertical growth, where increased cold wire feeding enhances deposition volume and promotes taller, more uniform layers. Conversely, excessive current flattens the molten pool, reducing effective buildup. The regression model developed in this work (Ht = 26.02-0.0239 I + 0.712 TS + 1.990 WR) quantitatively links these parameters

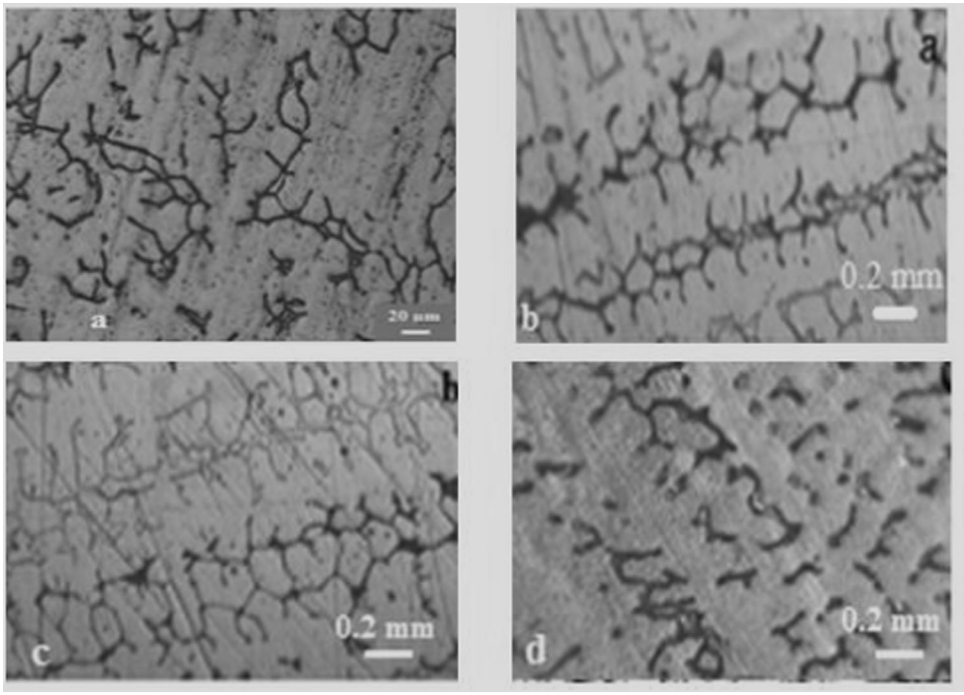


Fig. 7 Optical micro-photos show the effect of W.R on the microstructure: (a) without cold wire feeding, (b, c) low cold wire feeding, and (d) high cold wire feeding

Table 6 Confirmation test

I, A	TS, mm/sec	WR, m/min	Height, mm	Predicted, %	Error, %
160	4	2.5	30.79	30	2.5

to layer height, enabling predictive control of wall geometry within the tested range. Confirmation trials validated the model, achieving a maximum wall height of 30.79 mm with only 2.5% deviation from predicted values (Table 6). Beyond geometric considerations, layer height also influences microstructural development: specimens with higher, stable layer heights exhibited vertically aligned dendritic colonies and refined granular matrices, while insufficient height led to coarser, less uniform morphologies (Fig. 7). Thus, additively deposited layer height is not merely a dimensional metric but a critical indicator of process stability, deposition efficiency, and microstructural integrity. A detailed understanding of this concept provides a foundation for optimizing WAAM parameters to achieve defect-free, mechanically robust stainless-steel components.

4. Conclusions

This study systematically investigated the effects of travel speed, welding current, and wire feed ratio on the geometric and microstructural properties of SS309L stainless-steel layers fabricated using robotic double-wire MIG WAAM. The Taguchi L9 orthogonal array and ANOVA analysis revealed that wire feed ratio is the most significant parameter, contributing 43.6% to wall height variation, followed by travel speed (22.3%) and welding current (15.9%). The developed linear regression model accurately predicted wall height, with confirmation experiments showing a deviation below 2.5%.

Microstructural analysis demonstrated that cold wire feeding promotes the formation of vertically aligned dendritic structures and a refined granular matrix, resulting in dense, defect-free deposits and improved mechanical integrity. In contrast, specimens produced without cold wire feeding exhibited coarser dendritic features and less uniform morphology.

Overall, the findings provide practical guidance for optimizing WAAM process parameters to achieve precise geometries and superior microstructural characteristics in stainless-steel components. This work supports the advancement of additive manufacturing technologies for industrial applications, emphasizing the importance of wire feed ratio and travel speed in achieving high-quality, reliable deposits.

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